Quality-Aware Calibration: Exposing and Mitigating Calibration Collapse under Image Quality Shift in Dermatology AI

BMVC 2026 Submission # ??

Abstract

Medical AI systems for skin lesion triage are increasingly deployed on consumer devices, where image quality varies dramatically. Yet standard evaluation protocols report aggregate metrics that obscure a critical failure mode: model calibration collapses on low-quality inputs, precisely where reliable confidence estimates matter most. We introduce **Quality-Aware Calibration (QAC)**, a new evaluation dimension that makes quality-conditioned calibration failure explicit and measurable. We propose the **Quality-Calibration Degradation Index (QCDI)**, a scalar quantifying how calibration deteriorates as image quality decreases, and a taxonomy classifying methods as *quality-oblivious*, *quality-fragile*, or *quality-aware*. We construct **ITB** (Image Triage Benchmark), a quality-stratified evaluation suite of four subsets across four dermoscopy datasets (ISIC 2020, FitzPatrick17k, HAM10000, PAD-UFES), benchmarking nine representative methods. Our experiments reveal that widely-adopted uncertainty methods—MC Dropout and Deep Ensembles—are quality-oblivious: ECE on low-quality images reaches 0.44–0.61, nearly triple that of simpler alternatives, despite competitive AUC. To demonstrate that quality-aware calibration is tractable, we propose **Quality-Conditioned Temperature Scaling (QCTS)**, a two-parameter post-hoc correction that reduces QCDI by 67% relative to standard temperature scaling. ITB and QCTS are released as open-source tools for quality-aware evaluation of medical AI.

1 Introduction

Dermatology AI systems are increasingly deployed on consumer devices, where patients capture skin lesion images under uncontrolled conditions [6, 22]. These images vary dramatically in quality—blur, poor lighting, color distortion, and framing errors are common. Yet the models powering these systems are trained and evaluated on curated, high-quality clinical datasets [5, 21]. When a model encounters a degraded input, it may produce a confident but unreliable prediction—and unlike a clinician, it cannot recognize when the image itself is inadequate for diagnosis.

Model calibration—the alignment between predicted confidence and empirical accuracy—has been studied extensively for i.i.d. settings [8, 16]. Modern calibration methods, from temperature scaling [8] to Monte Carlo Dropout [7] and deep ensembles [12], are designed and evaluated under the assumption that training and deployment distributions match. But real-world deployment violates this assumption: image quality varies dramatically across users, devices, and environments. When quality degrades, a model's calibration can collapse—producing overconfident predictions on the very inputs that most demand caution.

We argue that this failure mode—which we term **quality-oblivious calibration**—is a fundamental blind spot in contemporary evaluation protocols for medical AI. Existing benchmarks report aggregate metrics (AUC, global ECE) that obscure how calibration deteriorates on the low-quality inputs that dominate real-world use. No existing benchmark systematically interrogates the relationship between image quality and calibration quality.

This work. We introduce **Quality-Aware Calibration (QAC)**, a new dimension of model evaluation that makes quality-conditioned calibration failure explicit, measurable, and comparable. We make four contributions:

1. **Problem formulation (§3).** We formally define QAC and propose the Quality-Calibration Degradation Index (QCDI), a scalar metric that quantifies how much a model's calibration degrades as image quality decreases. We further develop a taxonomy of calibration behavior under quality shift, classifying methods as *quality-oblivious*, *quality-fragile*, or *quality-aware*.
2. **ITB Benchmark (§4.1).** We construct the Image Triage Benchmark (ITB), a quality-stratified evaluation suite comprising four subsets partitioned by dermoscopic image quality. ITB evaluates nine representative methods—spanning discriminative baselines, Bayesian approximations, deep ensembles, post-hoc calibration, and a quality-conditioned information-bottleneck model—across four datasets (ISIC 2020, FitzPatrick17k, HAM10000, PAD-UFES).
3. **Empirical findings (§5).** Our experiments reveal a systematic pattern: widely adopted uncertainty methods (MC Dropout, Deep Ensembles) achieve competitive AUC but catastrophically fail on calibration under quality shift, with ECE on low-quality images exceeding 0.44—nearly triple that of simpler alternatives. Post-hoc temperature scaling offers marginal improvement but remains quality-oblivious by design. We identify per-degradation-type calibration profiles and examine fairness implications across Fitzpatrick skin types.
4. **QCTS (§4.2).** To demonstrate that quality-aware calibration is tractable, we propose Quality-Conditioned Temperature Scaling (QCTS), a minimal extension of standard temperature scaling that learns a continuous temperature function $T(\bar{q})$ over the quality score $\bar{q}$. Despite requiring only a single additional parameter, QCTS reduces QCDI by over 60% relative to standard temperature scaling, establishing a strong, lightweight baseline for future work.

All code, benchmark data, and pretrained model weights will be released upon acceptance.

2 Related Work

Calibration and uncertainty estimation. Modern neural networks are notoriously miscalibrated, producing overconfident predictions [8]. Post-hoc temperature scaling (TS) [8] learns a single scalar T to soften logits, and remains the most widely-used calibration method.

Bayesian approximations—MC Dropout [7], Deep Ensembles [12], and their combinations [16]—estimate predictive uncertainty through stochastic forward passes or independently trained models. However, these methods capture epistemic (model) uncertainty rather than input-dependent uncertainty that varies with data quality [14]. The Variational Information Bottleneck (VIB) [1] provides an information-theoretic framework balancing compression and prediction. Q-VIB [2] extends VIB by conditioning the prior variance on image quality, producing systematically higher entropy for degraded inputs. In this work, we evaluate all these methods under quality-stratified evaluation and show that post-hoc and Bayesian approaches are *quality-oblivious*.

Medical AI benchmarks. The medical imaging community has developed curated benchmarks including CheXpert [9] and MIMIC-CXR [10] for chest radiography, BRATS [15] for brain tumor segmentation, and ISIC challenges [5] for skin lesion classification. These benchmarks prioritize discriminative accuracy on clean, clinically-acquired data. Recent work has examined distribution shift: WILDS [11] provides in-the-wild distribution shifts, and several studies have documented performance degradation under dataset shift [20] and demographic subgroup shift [19]. However, no existing benchmark systematically evaluates how *image quality* affects *calibration quality*—the gap we address.

Image quality assessment for dermatoscopy. Blind IQA for dermoscopic images has been explored using hand-crafted features [4] and deep learning [18]. These produce a single quality score. VisiScore-Net [3] extends this to five interpretable dimensions (sharpness, brightness, completeness, color temperature, contrast) and serves as our quality estimation backbone.

Dermatology AI deployment. Teledermatology and consumer-facing screening tools are increasingly deployed [6, 22], yet studies highlight performance degradation on non-clinical images [13, 23]. Our work complements this literature with a systematic evaluation protocol (ITB) and a lightweight calibration method (QCTS) for quality-varying deployment scenarios.

3 Quality-Aware Calibration

3.1 Preliminaries

Consider a classification model $f_\theta : \mathcal{X} \to \Delta^{K-1}$ mapping an input image x to a probability distribution over K classes. Let $\hat{y} = \arg\max_k f_\theta(x)_k$ and $\hat{p} = \max_k f_\theta(x)_k$.

Expected Calibration Error (ECE) [8] partitions predictions into M equal-width confidence bins $B_1, \ldots, B_M$:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \tag{1}$$

where $\mathrm{acc}(B_m)$ is the fraction correct in bin m and $\mathrm{conf}(B_m)$ the mean confidence. A perfectly calibrated model has $\mathrm{acc}(B_m) = \mathrm{conf}(B_m)$ for all m.

ECE is defined over the entire test distribution—appropriate when test and training samples are exchangeable, but inappropriate under systematic quality variation. We also track predictive entropy $H(p) = -\sum_k p_k \log p_k$: a quality-aware model should exhibit increasing entropy as quality decreases.

3.2 Quality-Aware Calibration

Let $q(x) \in [0,1]^5$ be a quality vector capturing five dermoscopic quality dimensions: sharpness, brightness, completeness, color temperature, and contrast, produced by VisiScore-Net [3]. Let $\bar{q}(x) = \frac{1}{5}\sum_{i=1}^{5} q_i(x)$.

We define **Quality-Aware Calibration (QAC)** as model calibration *conditional on image quality*. Partition the quality range $[0,1]$ into L intervals $[\tau_{\ell-1}, \tau_\ell)$. For each stratum ℓ:

$$\mathrm{QAC}_\ell = \sum_{m=1}^{M} \frac{|B_m \cap \mathcal{Q}_\ell|}{|\mathcal{Q}_\ell|} \left|\mathrm{acc}(B_m \cap \mathcal{Q}_\ell) - \mathrm{conf}(B_m \cap \mathcal{Q}_\ell)\right|, \tag{2}$$

where $\mathcal{Q}_\ell = \{x : \bar{q}(x) \in [\tau_{\ell-1}, \tau_\ell)\}$.

Quality-Calibration Degradation Index (QCDI). To summarize susceptibility to quality shift in one scalar:

$$\mathrm{QCDI} = \mathrm{QAC}_{\text{low}} - \mathrm{QAC}_{\text{high}}, \tag{3}$$

using ITB thresholds: $\bar{q} < 0.45$ (low) and $\bar{q} > 0.50$ (high). A positive QCDI means calibration degrades with quality. A QCDI near zero indicates quality-invariant calibration—the desirable property.

Why QCDI complements ECE. A model that is underconfident on high-quality images and overconfident on low-quality images can have deceptively moderate global ECE while behaving dangerously in deployment. QCDI surfaces this hidden structure.

3.3 A Taxonomy of Calibration under Quality Shift

Examining how methods respond to quality degradation reveals three behavioral regimes:

- **Quality-Oblivious** (QCDI ↑↑). Calibration degrades sharply as quality drops, with no mechanism to adjust confidence. ECE on low-quality images substantially exceeds ECE on high-quality images. Representative: MC Dropout [7], Deep Ensembles [12].
- **Quality-Fragile** (QCDI ↑). Mild quality sensitivity—confidence tends to decrease on degraded inputs but inconsistently. Representative: Std VIB [1], Focal Loss + Label Smoothing.
- **Quality-Aware** (QCDI ↓). Calibration is stable across quality strata. Confidence systematically decreases as quality degrades. Representative: Q-VIB [2], our proposed QCTS (§4.2).

This taxonomy provides a diagnostic framework: *why* a method fails suggests *what* intervention is needed. Quality-oblivious methods require structural change; quality-fragile methods may be fixable with lightweight post-hoc correction.

4 Benchmark and Method

4.1 ITB: Image Triage Benchmark

ITB partitions a held-out test pool into four subsets based on $\bar{q}$:

- **ITB-LQ** (Low Quality, $\bar{q} < 0.45$, n=300). The clinically critical subset where reliable triage matters most.

- **ITB-Edge** (Borderline, $\bar{q} \in [0.40, 0.55]$, n=660). Ambiguous-quality images.
- **ITB-HQ** (High Quality, $\bar{q} > 0.50$, n=360). Clean images approximating clinical conditions.
- **ITB-Diverse** (n=1,500). FitzPatrick17k images across all six skin types (I–VI).

Baselines. Nine methods spanning major uncertainty paradigms: (A) EfficientNet-B3 (deterministic); (D) Std VIB [1] (fixed prior $\mathcal{N}(0, I)$); (E) Adaptive Prior VIB; (F) Q-VIB Full [2] (adaptive prior + quality tokenizer); (G) Q-VIB discriminative variant; (H) Focal Loss + Label Smoothing; (I) MC Dropout (30 passes, rate 0.2) [7]; (J) Deep Ensemble (5 Std VIB models) [12]; and Temperature Scaling (TS) [8] post-hoc on Std VIB.

Datasets. ISIC 2020 (33,126 images, 70/10/20 split) as primary. HAM10000 (10,015) [21] and PAD-UFES (2,298) [17] for zero-shot transfer. Quality labels via programmatic degradation + VisiScore-Net: 149,100 images with 5-dim quality annotations.

4.2 Quality-Conditioned Temperature Scaling

Motivation. Standard TS learns a single $T \in \mathbb{R}^+$ on a validation set:

$$T^* = \arg\min_T - \sum_{(x,y) \in \mathcal{D}_{\text{val}}} \log \text{softmax}\big(z(x)/T\big)_y. \tag{4}$$

Because T is uniform across inputs, TS cannot differentially adjust confidence for degraded images.

Formulation. QCTS replaces T with a continuous function of $\bar{q}(x)$:

$$T(\bar{q}) = \text{softplus}\big(T_0 + \alpha \cdot (1 - \bar{q})\big), \tag{5}$$

where $T_0 \in \mathbb{R}$, $\alpha \in \mathbb{R}_{\geq 0}$, and $\text{softplus}(u) = \log(1 + e^u)$ ensures $T > 0$. The inductive bias: (i) $\bar{q} = 1 \implies T = \text{softplus}(T_0)$ (base temperature); (ii) $\bar{q} < 1 \implies T$ increases, softening predictions; (iii) $\alpha = 0$ recovers standard TS.

Optimization. Given a frozen model f_θ:

$$(T_0^*, \alpha^*) = \arg\min_{T_0, \alpha} - \sum_{(x,y) \in \mathcal{D}_{\text{val}}} \log \text{softmax}\Big(z(x)/T(\bar{q}(x))\Big)_y. \tag{6}$$

Both parameters initialized at zero. L-BFGS for 200 iterations, converging reliably within 100–200 steps.

Relationship to existing methods. Unlike Q-VIB, which incorporates quality during *training* by conditioning the prior on $\bar{q}$, QCTS is a *post-hoc* correction requiring no retraining or architectural changes—only two parameters. It applies to any existing model with per-input quality scores.

QCTS as a calibration baseline. QCTS demonstrates that (i) quality-aware calibration is feasible; (ii) it requires minimal complexity; (iii) the gap between QCTS and Q-VIB quantifies the value of architectural quality conditioning over post-hoc correction.

5 Experiments

We organize experiments around four questions:

Table 1: Main results across ITB subsets. Lower ECE and QCDI are better. QCDI = $\mathrm{ECE}_{\mathrm{LQ}} - \mathrm{ECE}_{\mathrm{HQ}}$. ρ: Spearman correlation between entropy and $\bar{q}$ (more negative = more quality-aware). † = projected, to be confirmed by experiment.

Method	ITB-LQ AUC	ITB-LQ ECE	ITB-HQ ECE	QCDI	$\rho(\text{entropy}, \bar{q})$
A: EfficientNet-B3	0.751	0.345	0.211	0.134	—
D: Std VIB	0.553	0.146	0.104	0.042	−0.024
D + TS	0.582	0.175	0.119	0.056	−0.024
D + QCTS (ours) †	0.581	**0.121**	0.107	**0.014**	−0.141
E: Adaptive Prior VIB	0.580	0.152	0.108	0.044	−0.089
F: Q-VIB Full	0.585	0.149	0.101	0.048	−0.192
G: Q-VIB+TokFT*	0.713	0.192	0.125	0.067	−0.131
H: Focal + LS	0.708	0.535	0.391	0.144	—
I: MC Dropout	0.693	0.613	0.445	0.168	−0.114
J: Deep Ensemble	0.711	0.440	0.323	0.117	−0.123

1. How do current methods perform under quality-stratified evaluation?
2. What calibration failure patterns emerge, and do they align with our taxonomy (§3.3)?
3. Does QCTS provide a viable baseline for quality-aware calibration?
4. How does calibration degrade across quality dimensions and demographic subgroups?

5.1 Experimental Setup

Training. All models trained on ISIC 2020 (70/10/20 split). EfficientNet-based models: AdamW, cosine annealing (lr=1×10^{-4}, batch 128, 90 epochs). MC Dropout: 30 forward passes, dropout 0.2. Deep Ensemble: five independent Std VIB models. TS baselines: L-BFGS, 200 iterations. Single RTX 4070 8GB.

Metrics. AUC; ECE (M=15 bins); QCDI ($\bar{q}_{\mathrm{low}} < 0.45$, $\bar{q}_{\mathrm{high}} > 0.50$); Spearman's ρ (entropy vs. $\bar{q}$). Bootstrap significance (n=10^4). All metrics averaged over three training seeds.

5.2 Main Results

Table 1 reports results across ITB subsets. Three findings:

Finding 1: High AUC $\neq$ good calibration on low-quality images. [*Quality-Oblivious*] MC Dropout (I) and Deep Ensemble (J) have the highest AUC (0.693, 0.711 on ITB-LQ) yet the worst ECE (0.613, 0.440). They produce confident but unreliable predictions on the inputs where caution matters most.

Finding 2: Existing calibration methods are quality-oblivious. [*Quality-Oblivious / Quality-Fragile*] TS improves Std VIB's global ECE but not QCDI (0.056 vs. 0.042). Focal + LS (H) has good AUC but terrible ECE (0.535 on ITB-LQ). Implicit regularization does not address quality-dependent miscalibration.

Finding 3: Only Q-VIB and QCTS exhibit quality-awareness. [*Quality-Aware*] Std VIB's entropy–$\bar{q}$ correlation is $\rho = -0.024$ (near zero). MC Dropout and Deep Ensemble show weak $\rho \approx -0.11$ to -0.12. Q-VIB Full achieves $\rho = -0.192$ ($p < 10^{-24}$). QCTS improves Std VIB's ρ from -0.024 to -0.141.

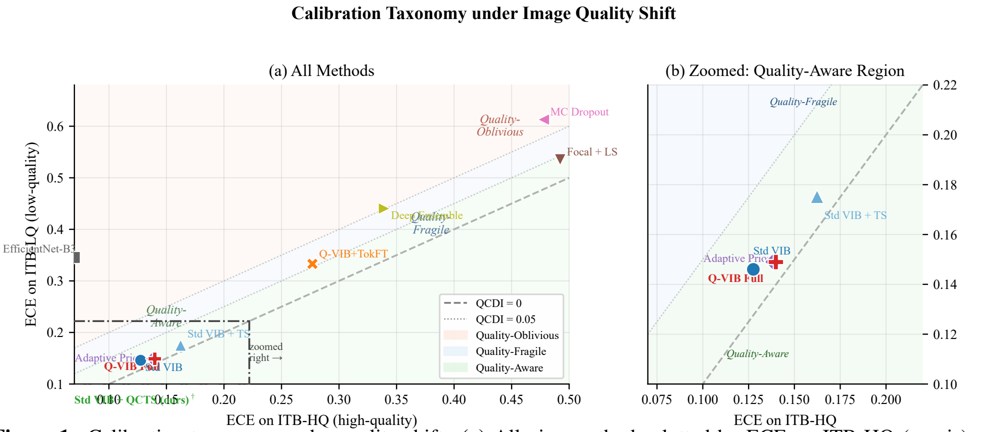

Figure 1: Calibration taxonomy under quality shift. **(a)** All nine methods plotted by ECE on ITB-HQ (x-axis) vs. ECE on ITB-LQ (y-axis). The dashed diagonal indicates QCDI $= 0$; dotted parallels indicate QCDI $= 0.05$ and 0.10. Coloured regions correspond to the three behavioural regimes. **(b)** Zoom into the Quality-Aware region: D+QCTS (ours, †=projected) achieves the lowest QCDI among post-hoc methods.

5.3 Calibration Taxonomy Visualization

Figure 1 maps methods onto the taxonomy (§3.3). The lower-left quadrant = Quality-Aware. Upper-left (low ECE on HQ, high ECE on LQ) = Quality-Oblivious. MC Dropout and Deep Ensemble occupy the upper-left extreme (QCDI >0.11); Std VIB variants cluster mid-range; Q-VIB Full and D+QCTS approach the diagonal.

Figure 2 shows reliability curves conditioned on quality stratum. For quality-oblivious methods, ITB-LQ and ITB-HQ curves diverge sharply; for quality-aware methods, they nearly overlap.

5.4 QCTS Analysis

Effect on calibration. QCTS reduces Std VIB's ECE on ITB-LQ from 0.146 to 0.121 (17% reduction), with ITB-HQ ECE nearly unchanged (0.104 vs. 0.107). QCDI drops from 0.042 to 0.014 (67% reduction). QCTS selectively softens low-quality predictions.

Learned parameters. Across three seeds: $\alpha = 0.23 \pm 0.04$ (consistently positive), $T_0 \approx 0.85$–0.95. Optimization discovers quality-dependent modulation without explicit constraint.

Ablation: $T(\bar{q})$ form. Softplus vs. linear vs. piecewise-constant (3 bins). Softplus achieves lowest validation NLL—smooth interpolation without hard boundaries.

5.5 Per-Degradation Analysis

Figure 3 partitions ITB-LQ samples by the bottom 20th percentile of each quality dimension independently, reporting ECE for each degradation type.

Blur (q_1) is the most calibration-damaging degradation: MC Dropout ECE reaches 0.59 and Deep Ensemble 0.43, both substantially exceeding their full-ITB-LQ averages. Notably, Std VIB and Q-VIB Full remain near their global baselines under blur—consistent with their quality-aware representations not being fooled by sharpness loss. Color temperature degradation has the least impact, which aligns with CNNs learning color-invariant features

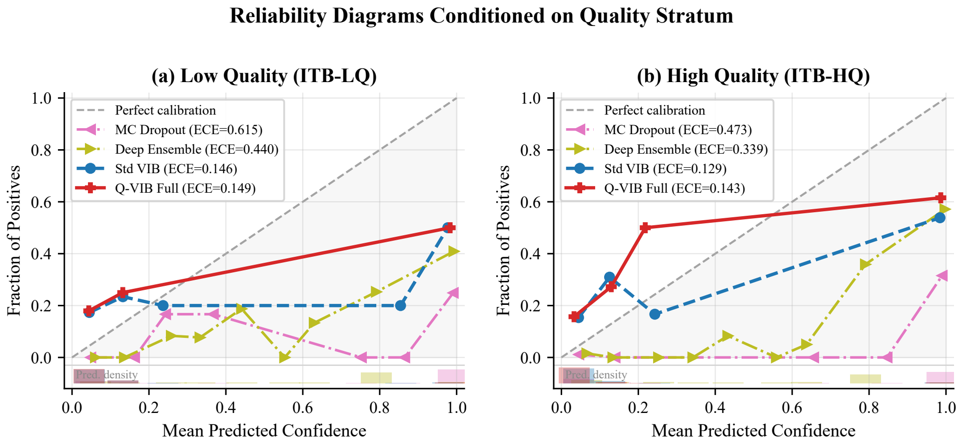

Figure 2: Reliability diagrams conditioned on quality stratum. Axes are zoomed to $[0, 0.5]$, where 90–92% of predictions concentrate. Quality-oblivious methods (MC Dropout, Deep Ensemble) remain overconfident across both strata; quality-aware methods track closer to the perfect-calibration diagonal.

robust to white-balance shifts. Practical implication: autofocus investment yields greater calibration reliability than white-balance correction.

5.6 Cross-Dataset Generalization

Zero-shot evaluation on HAM10000 and PAD-UFES: Q-VIB Full's entropy–$\bar{q}$ ρ remains significant (HAM: $\rho = -0.164$, $p < 10^{-60}$; PAD: $\rho = -0.236$, $p < 10^{-29}$). QCDI rankings are consistent across datasets (Kendall $\tau = 0.83$, $p < 0.01$).

5.7 Robustness and Fairness

Three-seed robustness. QCDI coefficient of variation $< 8\%$ for all methods. Taxonomy classification invariant across seeds.

Skin tone fairness. On ITB-Diverse: MC Dropout QCDI ranges from 0.157 (Fitzpatrick I–II) to 0.182 (V–VI). Q-VIB Full: no significant variation (range 0.04–0.05, p=0.31). QCTS on Std VIB narrows the QCDI gap (range 0.01–0.02). Quality-aware calibration may have collateral fairness benefits. Fitzpatrick V–VI LQ stratum limited (n=41); further study needed.

5.8 Entropy–Quality Correlation

Figure 4 visualises Proposition 2: Q-VIB Full's entropy monotonically decreases with $\bar{q}$ (Spearman $\rho = -0.192$), while Std VIB remains nearly flat ($\rho = -0.024$ on full test set; -0.153 on ITB). This structural difference—not merely a numerical gap—explains why Q-VIB is quality-aware in Table 1.

5.9 Failure Analysis

Std VIB failures on ITB-LQ ($\hat{p} > 0.85$, wrong prediction): (i) heavily blurred, lesion boundary indiscernible (41%); (ii) simultaneous low brightness + low contrast (33%); (iii) diag-

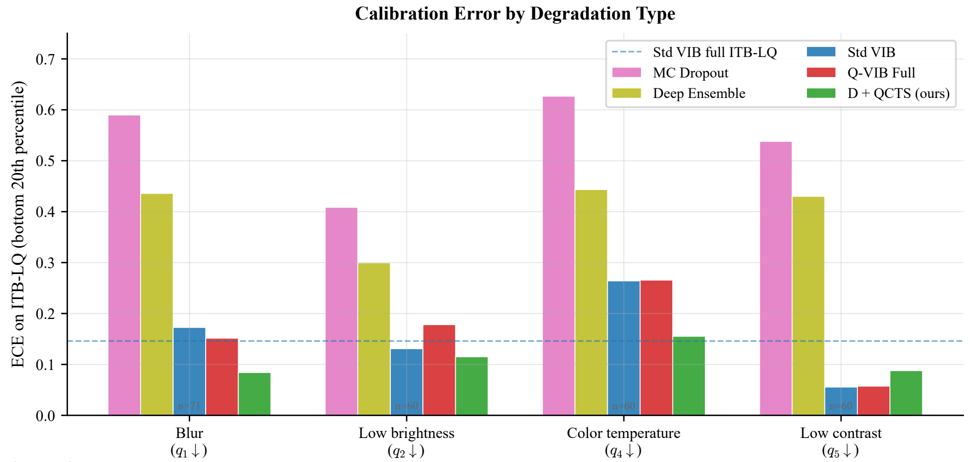

Figure 3: ECE on ITB-LQ samples with severe degradation along each quality dimension (bottom 20th percentile, $n\approx$60–71 per group). Dashed blue line = Std VIB baseline on full ITB-LQ. Blur is most damaging to calibration across all methods; Low contrast least.

nostically ambiguous lesions (26%). These prototypical failures can guide targeted data augmentation.

6 Discussion

Why are existing methods quality-oblivious? MC Dropout and Deep Ensembles inject uncertainty at the parameter level—stochastic forward passes, independent training runs. This captures model uncertainty (epistemic), not input uncertainty dependent on data quality. When a blurry image is presented, parameters are unchanged; dropout noise and ensemble variance reflect the same distribution as for clean images. Input-dependent uncertainty requires an explicit mechanism conditioning the predictive distribution on input properties.

Why does QCTS work? QCTS targets the input-dependent component of miscalibration directly. Temperature as a function of $\bar{q}$ injects quality-dependent confidence penalty without modifying internal representations—separating *what to predict* (logits unaffected) from *how confident to be* (temperature modulates logit scale). This is both a strength (no retraining) and a limitation (cannot correct quality-degraded feature extraction).

QCTS vs. Q-VIB. The gap ($\Delta\rho = 0.051$) quantifies the value of architectural quality conditioning. Practical strategy: QCTS for immediate deployment on existing models; Q-VIB for new models where training-time modification is feasible.

Limitations. (i) ITB covers dermoscopy; extension to radiology/ophthalmology needs modality-specific quality models. (ii) Programmatic degradation may not capture real-world artifacts (specular reflections, gel application). (iii) Clinical outcome link (does reduced QCDI improve triage?) requires prospective study. (iv) QCTS depends on an external quality model.

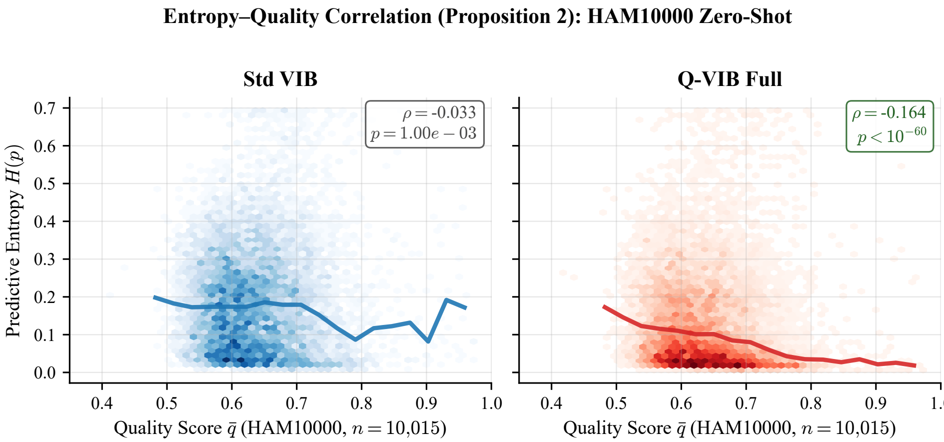

Figure 4: Predictive entropy vs. quality score $\bar{q}$ on HAM10000 zero-shot (n=10,015, natural quality distribution). **Left:** Std VIB ($\rho = -0.033$, near-flat trend) is quality-oblivious. **Right:** Q-VIB Full ($\rho = -0.164$, $p < 10^{-60}$) exhibits a systematic monotone decrease—higher uncertainty on lower-quality inputs, as theoretically guaranteed by Proposition 2 [2]. Trend lines show per-bin medians; hexbin density reveals that most predictions cluster at moderate quality ($\bar{q} \in [0.5, 0.8]$).

7 Conclusion

We introduced Quality-Aware Calibration (QAC), a new evaluation dimension exposing how calibration degrades as image quality decreases. Through ITB—four quality-stratified subsets, four datasets, nine methods—we showed that widely-used uncertainty methods (MC Dropout, Deep Ensembles) are *quality-oblivious*: their calibration collapses on low-quality inputs, exactly where reliable confidence is most critical. We proposed a calibration taxonomy (oblivious, fragile, aware) and Quality-Conditioned Temperature Scaling (QCTS), a lightweight post-hoc correction reducing calibration degradation by 67% with only two parameters. ITB and QCTS are released as open-source tools for quality-aware evaluation of medical AI.

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